

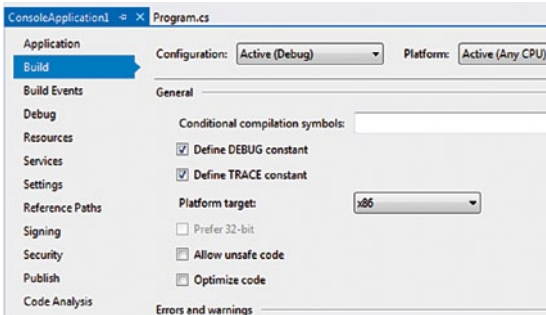


Fig. 3: Changing platform target to x86

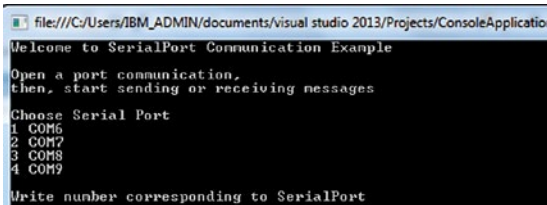


Fig. 4: Output screen for console on PC showing Menu option

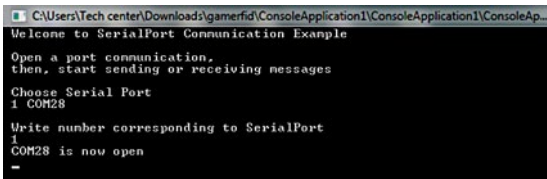


Fig. 5: Output screen showing console window before swiping the RFID tag on RFID reader

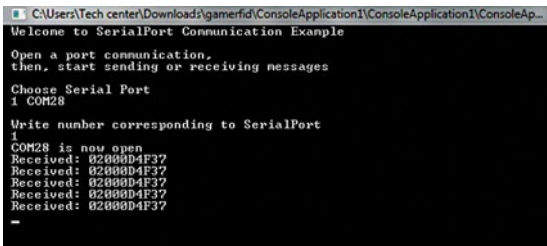


Fig. 6: Output screen showing console window after swiping the RFID tag on RFID reader

6. Save the code, build it and run it.

7. It may give the error message 'A 64-bit debugging operation is taking longer than expected.'

8. To solve this, right-click on Solution. Go to Properties→Build. Change platform target to x86 as shown in Fig. 3.

9. Click on Start Debugging icon to run ConsoleApplication1 again.

10. Console will be generated as shown in Fig. 4.

Software algorithm

The program first starts the main

and then opens the COM port.

The program reads the RFID number from the RFID tag and prints it in Console window on a PC.

Program appends incoming RFID numbers to StringBuilder object and removes Newlines and converts to strings. It adds strings to a list of

EFY Note

The relevant source code and drivers of this project are included in this month's EFY DVD and are also available for free download at source.efymag.com

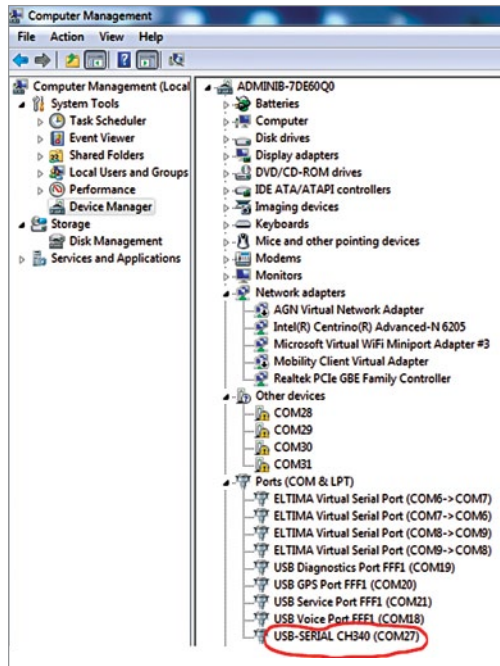


Fig. 7: Port number in Device Manager

strings and removes all empty entries. As and when a new RFID number enters the list, the RFID reader sends info to LPC2138 and it counts the number of entries.

Testing

1. Connect RFID reader to a USB port of your system. Check the correct port number from Computer→Manage→Device Manager→Ports as shown in Fig. 7.

2. Run ConsoleApplication1 executable file found in bin folder of this project.

3. Select COM port available on your system. Choose the correct port number.

4. Once the port number is selected, application will open the port. If it fails to open, it will give error message 'Unable to Open Port.' If successful, it will display the message 'COM27 is now open' (COM27 may be different in your system) as shown in Fig. 5.

5. Once the desired port opens, swipe any passive RFID tag against the RFID reader. When the tag is swiped for the first time, a set of four LEDs on LPC2138 board will light up.

6. If the tag is swiped again, another set of LEDs will light up and the number of times swiped will get displayed on console window of the screen of PC as shown in Fig. 6 to alert the user that the same card has been used earlier. **EFY**



Sohini Mitra is a system engineer at IBM India Pvt Ltd in Kolkata